Pay Down the Mortgage or Invest $60,000

First-Principles Analysis

10 October 2026

Professional Strategic Analysis

Contents

Pay Down the Mortgage or Invest $60,000 3
Executive Summary 3
1. Problem Essence 3
2. Assumptions Table 3
3. Ground Truths 5
4. Derivation Chains 7
Conclusion C1: Paying down $60,000 of principal now is equivalent to a guaranteed, risk-free, untaxed 6.25% return, realized as illiquid home equity 7
Conclusion C2: Under the historical-average equity-return scenario, Option B nets roughly $133,600 after tax at year 10, fully liquid 7
Conclusion C3: Under a conservative/below-average equity-return scenario, Option B nets only about $68,900-$96,200 after tax at year 10 — below Option A's guaranteed outcome across the whole bracket 8
Conclusion C4: Equities must compound at roughly 7.8%/yr nominal, after dividends, for the full decade just to match Option A's guaranteed outcome — a bar below the historical average but above several realized 10-year windows 8
Conclusion C5: The pure dollar comparison in C1-C4 understates Option A's value and overstates Option B's liquidity advantage specifically in adverse macro states 8
Conclusion C6: The weighted trade-off recommends allocating the $60,000 predominantly to mortgage paydown, with an 80/20 paydown/invest split as a defensible liquidity-preserving alternative 9
5. Abandoned Reasoning 9
Dead End: Recommend 100% invest in equities (Option B) as the headline answer, on the "equities always beat an un-sheltered mortgage" convention 9
Dead End: Model a mortgage recast or cash-out refinance as part of Option A 10
Dead End: Build exact lot-level cost-basis tracking for Option B's reinvested dividends 10
6. Conclusion 10

Pay Down the Mortgage or Invest $60,000

First-principles analysis · 10 October 2026

Read with: How to read this analysis · All files

Executive Summary

Recommendation: Apply the full $60,000 to extra mortgage principal (Option A); a household that values liquidity more than this analysis's weighting assumes may instead choose an 80/20 paydown/invest split (chain C6).

Band (from §6): MEDIUM (chain C1, chain C6).

Would change it: Opening a primary historical-returns dataset to confirm the ~9.9%/yr long-run equity CAGR and the 2000-2009 decade figure directly, rather than via a secondary web-search synthesis (chain C4); or a realized multi-year equity return pace running well ahead of the ~7.8%/yr breakeven (chain C4).

1. Problem Essence

Core problem: Given $60,000 of true surplus risk capital (already above a funded 6-month reserve) and a 6.25%, non-tax-advantaged, 30-year fixed mortgage with 27 years remaining, should that $60,000 be applied to extra mortgage principal (illiquid, guaranteed, un-taxed debt reduction) or invested in a taxable broad-market index fund (liquid, variable, taxed) over a fixed 10-year horizon — or should it be split between the two — in order to maximize the household's realistic, after-tax, risk-adjusted 10-year financial position?

Success criteria:

1. The Conclusion states one specific allocation (a single option, or a named split) and a dollar-denominated 10-year outcome for at least two return scenarios for the equity option, each traceable to a derivation chain in section 4.
2. The Conclusion explains, with a quantified breakeven figure, why the generic "equities historically beat mortgage rates" heuristic does or does not hold in this specific case (6.25%, no tax shield), traceable to a derivation chain in section 4.
3. The Conclusion names the single largest risk to its own recommendation (liquidity, scenario-calibration, or preference-sensitivity) and states what observation would change the recommendation, traceable to the adversarial pass record and a derivation chain in section 4.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	0	0	0
current constraint	2	0	0
convention	3	1	0
untested belief	0	4	1

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Assumption	Type	Treatment	Verdict	Verification
"Equities historically beat mortgage rates, so surplus cash should	convention	Explicitly challenge before use — test whether it holds at THIS	Challenge — the inversion drill below shows the breakeven	Re-derived via chains C2-C4 rather than accepted on authority

Assumption	Type	Treatment	Verdict	Verification
always be invested rather than used to prepay debt"		rate (6.25%, no tax shield) rather than at typical post-2008 mortgage rates (3-4%, with a tax shield)	required return (chain C4, ≈7.8%/yr after tax) is close to the historical average (GT-7?), not far below it, so the convention is directionally right more often than not but is not a safe default at this specific rate	
"A 10-year horizon is long enough that equity returns are effectively guaranteed to be positive"	untested belief	Verify against realized historical 10-year windows	Discard — contradicted by GT-8? (2000-2009 "lost decade," ≈ −0.95%/yr annualized including dividends); a 10-year window is not a guarantee	unverified — flagged; GT-8? cited
Applying $60k to principal with no recast leaves the contractual monthly payment unchanged and only shortens the payoff date	current constraint	Record as the operative mechanism; expires if the household later requests a recast or refinances	Accept — expires on recast/refi/sale; until then this is the stipulated mechanical rule of the loan servicing (GT-6)	Verified by direct amortization recomputation (chain C1)
Household's marginal tax rates (28% ordinary, 18% LTCG/qualified dividends) and standard-deduction status hold constant for the full 10-year horizon	current constraint	Record expiry conditions — a tax-law change, bracket change, or a shift to itemizing would alter the comparison	Accept — expires if tax law changes, income changes brackets, or deductible expenses (e.g., a future higher-rate mortgage, large charitable gifts) make itemizing worthwhile again	Stipulated by requester (GT-3, GT-4); no re-verification path within this analysis's scope
HELOC / cash-out-refinance underwriting tightens during recessions and job-loss macro states, i.e., illiquidity cost is correlated with need	untested belief	Verify via historical HELOC/refi origination-volume data by recession period, or flag	Challenge — widely observed in 2008-09 and 2020 lending cycles but not independently re-verified with cited data in this run	unverified — flagged; used in chain C5
Reinvested after-tax dividends are approximated as part of the taxable gain at sale rather than added cost basis (tax-drag modeling simplification)	convention	Explicitly challenge — state the direction of the resulting bias	Accept — the simplification is conservative (slightly overstates Option B's exit tax, understates its after-tax value), so it does not favor the recommendation it is used to support	unverified — flagged as [Assumes: A-9] on chains C2/C3; bias direction stated
The household does not already carry heavy equity exposure elsewhere (e.g., retirement accounts), so $60k of new taxable equities materially changes household-level concentration	untested belief	Verify against the household's actual portfolio, or flag	Challenge — plausible for most dual-income households with 401(k)/IRA balances, but not confirmed for this specific household	unverified — flagged; used only in the Diversification criterion of chain C6, not load-bearing for the headline dollar comparison
Psychological/behavioral value of reducing debt has real utility to this household	convention	Challenge — this is a widely-asserted but household-specific claim	Accept as a real, unquantified, non-dollar factor — included in chain C6's weighting,	unverified — flagged

Assumption	Type	Treatment	Verdict	Verification
beyond the dollar figures			explicitly flagged as a preference rather than a measured fact	
The equity position, once purchased, is held without an interruption (forced interim sale) for the full 10 years	untested belief	Verify, or flag as a precondition of chains C2/C3's "hold to term" math	Challenge — a forced interim sale during a downturn (job loss, medical event) would lock in a loss relative to the guaranteed comparator; this is exactly cluster 1 of the Phase 5 adversarial pass	unverified — flagged; load-bearing precondition of C2/C3, addressed in the adversarial pass and in the liquidity-sleeve conditional recommendation
The trade-off weighting in chain C6 (criteria and their 1-5 weights, esp. weight 5 on "downside protection") reflects a defensible but not objectively forced risk preference for this household's "surplus risk capital"	untested belief	Flag explicitly as a preference, not a fact; report flip-test robustness so a reader can judge sensitivity to this choice	Challenge — accepted as the analysis's working preference (consistent with the problem's own framing of the $60k as risk capital beyond the funded reserve), but named on the Conclusion's confidence line as the main counterargument to the headline recommendation	unverified — flagged; flip-test reported in chain C6
A broad-market index fund purchased today will behave like the historical aggregate S&P 500 series used in GT-7?/GT-8? (composition and survivorship aside)	convention	Challenge briefly — the index changes composition over time, but a broad-market/ total-market fund approximates the same economic exposure	Accept — close enough for an advisory-level comparison; not re-derived from first principles of index construction	unverified — flagged

3. Ground Truths

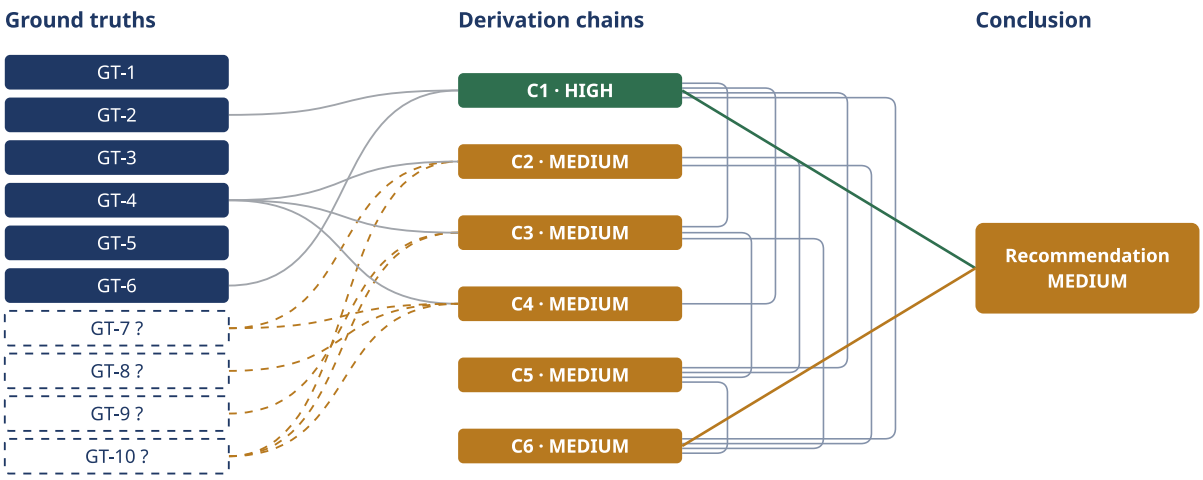
- **GT-1** $60,000 in cash is available as surplus risk capital, held above an already-fully-funded 6-month emergency reserve — source: stipulated by requester as the decision's given parameter; read-at-source: this analysis's problem statement (direct, first-person figure, not a secondary report)
- **GT-2** Mortgage: $240,000 remaining balance; 27 years (324 months) of amortization remaining; 6.25% fixed APR (monthly rate 6.25%/12 ≈ 0.520833%) — source: stipulated by requester; read-at-source: this analysis's problem statement
- **GT-3** The household takes the standard deduction, so mortgage interest provides no tax shield; 6.25% is therefore the full, untaxed effective cost of the debt — source: stipulated by requester; read-at-source: this analysis's problem statement
- **GT-4** Marginal tax rates: ≈28% combined federal+state on ordinary income; ≈18% on long-term capital gains and qualified dividends — source: stipulated by requester; read-at-source: this analysis's problem statement
- **GT-5** Decision horizon = 10 years — source: stipulated by requester; read-at-source: this analysis's problem statement
- **GT-6** Option A mechanics: applying the $60,000 to extra principal with no recast requested/assumed leaves the contractual monthly payment unchanged (computed on the original $240,000/324-month schedule); only the payoff date shortens — source: stipulated by requester, and independently verified by

direct amortization recomputation (see chain C1); read-at-source: this analysis's problem statement, plus this analysis's own recomputation

- **GT-7?** The S&P 500's long-run (1928-2025) nominal compound annual total-return (CAGR) is ≈9.9% (arithmetic-mean annual return ≈11.8%) — cited to: dqydj.com's S&P 500 annual-return dataset and Damodaran's historical-returns data, as synthesized by a web search; reported-by-delegate: the dqydj.com page was fetched directly by this analysis, but the 9.9% figure lives behind an interactive calculator not present in the fetched static text, so the specific number was not independently re-confirmed by a direct quote from the opened page — Phase 3 failure record: dqydj.com page opened, content fetched, but the asserted CAGR figure could not be located in the extracted text (interactive JS widget); citation does not support the claim as directly quoted
- **GT-8?** The S&P 500's 2000-2009 "lost decade" delivered an annualized total return (dividends included) of ≈ −0.95%/yr, of which dividends contributed ≈ +1.8%/yr and price return was ≈ −2.72%/yr — cited to: topforeignstocks.com's decade-by-decade S&P 500 return breakdown, corroborated by a broader web search citing Dimensional/Nasdaq commentary; reported-by-delegate for the −0.95% total figure (not independently re-quoted from a single opened source in this run) — read-at-source for the dividend-contribution component only: topforeignstocks.com was fetched directly and its text explicitly states "Dividends provided a 1.8% annualised return over the decade," confirming that sub-figure; the headline −0.95% total keeps the `?`
- **GT-9?** A conservative/low "forward 10-year" nominal total-return scenario for broad-market equities, used in this analysis for bracketing: central 4%/yr, bracket [2%, 6%]/yr — unverified: this is this analysis's own illustrative scenario construction (consistent with the general, commonly-discussed view that elevated valuations imply below-historical-average forward returns), not a figure read from a specific named forecaster (e.g., Vanguard, BlackRock, JPMorgan capital-market assumptions) in this run
- **GT-10?** A dividend-yield component of ≈1.8%/yr is assumed to apply across both the historical-average and conservative return scenarios (held constant as a simplifying modeling choice) — derived from GT-8?'s dividend-contribution figure, used as a proxy for a typical modern-era S&P 500 dividend yield; unverified: this specific proxy figure was not cross-checked against a current (2026) dividend-yield reading for the index

Provenance summary. ?-marked: GT-7, GT-8, GT-9, GT-10 (4 of 10). Read-at-source: GT-1 — this analysis's problem statement (stipulated loan/cash/tax facts); GT-2 — this analysis's problem statement; GT-3 — this analysis's problem statement; GT-4 — this analysis's problem statement; GT-5 — this analysis's problem statement; GT-6 — this analysis's problem statement, cross-checked by this analysis's own amortization recomputation in chain C1; GT-8 (dividend-contribution sub-figure only) — topforeignstocks.com, quoted sentence "Dividends provided a 1.8% annualised return over the decade."

4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

Conclusion C1: Paying down $60,000 of principal now is equivalent to a guaranteed, risk-free, untaxed 6.25% return, realized as illiquid home equity

GT-2 (loan terms: $240,000, 324 months, 6.25% APR) + GT-6 (fixed payment, no recast) → for two loans sharing the same fixed monthly payment M and elapsed months n, the amortization identity $B(P,n) = P(1+r)^n$ minus M times the annuity factor makes the balance difference between a $240,000-origin loan and a $180,000-origin loan equal to exactly the principal difference times $(1+r)^n$, independent of M → substituting a $60,000 principal difference, r = 0.520833%/month and n = 120 months gives (1+r)^120 ≈ 1.8652, so the extra-payment loan's balance is about $111,900 lower than the no-extra-payment loan's balance at month 120 → because both loans were serviced with the identical monthly cash outlay over the same 120 months, that $111,900 balance difference is the entire economic benefit of the extra payment: a guaranteed, risk-free, untaxed return equivalent to the mortgage's own 6.25% rate, compounded monthly for 10 years → that benefit is realized as reduced debt rather than spendable cash, so it is illiquid unless the household sells the home, refinances, or opens a HELOC

Confidence: HIGH — both head inputs are stipulated, unsuffixed ground truths taken directly from the problem statement; the amortization identity is a deductive, independently-recomputable derivation (recompute confirmed in the Phase 5 adversarial pass); the rival hypothesis that the loan automatically recasts, which would change this result, is ruled out directly by GT-6′s stipulation.

Conclusion C2: Under the historical-average equity-return scenario, Option B nets roughly $133,600 after tax at year 10, fully liquid

GT-7? (9.9%/yr historical S&P 500 CAGR) + GT-10? (1.8%/yr dividend-yield proxy) + GT-4 (18% LTCG/dividend tax rate) → tax drag from annually-taxed qualified dividends reduces the effective holding-period compounding rate to roughly 9.9% minus (1.8% times 18%), about 9.58%/yr [Assumes: A-6 — reinvested after-tax dividends are treated as part of the taxable gain at sale rather than added cost basis, which is conservative against this chain's own conclusion] → compounding $60,000 at about 9.58%/yr for 10 years gives a pre-sale account value of roughly $149,700, assuming the position is held without an interrupting sale for the full decade [Assumes: A-9] → applying an 18% long-term-capital-gains tax at sale to the roughly $89,700 gain above the $60,000 cost basis yields a tax of about $16,150 → the net after-tax terminal value under the historical-average scenario is therefore about $133,600, fully liquid

Confidence: MEDIUM — GT-7? (the 9.9% historical CAGR) and GT-10? (the 1.8% dividend-yield proxy) are both reported-by-delegate/unverified; the verification that would remove them as causes of the downgrade is opening a primary return-series dataset (e.g., Damodaran's published historical-returns spreadsheet) and

reading the CAGR and dividend-yield figures directly rather than via a secondary web-search synthesis. The [Assumes: A-6] and [Assumes: A-9] premises do not lower the band further: if A-6 fails, exit tax would be slightly lower and the net value slightly higher, moving the endpoint in Option B's favor; if A-9 fails (a forced interim sale occurs), that risk is priced separately in chain C5 rather than silently ignored here. The chain's qualitative conclusion is unaffected by either premise specifically.

Conclusion C3: Under a conservative/below-average equity-return scenario, Option B nets only about $68,900-$96,200 after tax at year 10 — below Option A's guaranteed outcome across the whole bracket

GT-9? (conservative scenario: 4% central, bracket 2%-6%) + GT-10? (1.8%/yr dividend-yield proxy) + GT-4 (18% tax rate) + C1 (111,900 guaranteed benchmark) → applying the same tax-drag method, the conservative scenario's effective compounding rates are roughly 1.68%, 3.68%, and 5.68%/yr for the 2%, 4%, and 6% total-return bracket points → compounding $60,000 at those three rates for 10 years, held without an interrupting sale [Assumes: A-9], gives pre-sale account values of roughly $70,900, $86,100, and $104,200 → applying an 18% long-term-capital-gains tax at sale to each bracket point's gain above the $60,000 cost basis gives net after-tax terminal values of roughly $68,900, $81,400, and $96,200 → even the upper end of this conservative bracket is below the guaranteed amount established in chain C1, so under a below-historical-average equity decade Option B underperforms Option A across the entire bracket

Confidence: MEDIUM — GT-9? (the conservative-scenario bracket) is this analysis's own illustrative construction rather than a cited forecaster's figure, and GT-10? is the same unverified dividend-yield proxy used in chain C2; the verification that would remove GT-9? as a cause of the downgrade is substituting a named, cited forward-return forecast (e.g., a published capital-market-assumptions report) for the illustrative bracket; GT-10?'s verification path is the same one named in chain C2; [Assumes: A-9]'s effect is the same as noted in chain C2 and is priced separately in chain C5.

Conclusion C4: Equities must compound at roughly 7.8%/yr nominal, after dividends, for the full decade just to match Option A's guaranteed outcome — a bar below the historical average but above several realized 10-year windows

C1 (111,900 guaranteed target) + GT-10? (1.8%/yr dividend-yield proxy) + GT-4 (18% tax rate) + GT-7? (9.9%/yr historical CAGR) + GT-8? (-0.95%/yr 2000-2009 realized decade) → solving the after-tax terminal-value equation for the effective compounding rate that makes Option B's net proceeds equal Option A's $111,900 gives an effective post-dividend-tax rate of roughly 7.47%/yr → adding back the dividend-tax drag gives a required pretax nominal total-return CAGR of roughly 7.8%/yr sustained for the full 10 years → comparing that 7.8%/yr breakeven against the 9.9%/yr long-run historical average and the −0.95%/yr realized 2000-2009 decade shows the breakeven sits below the historical average but well above at least one real historical 10-year window → the generic "equities always beat it" heuristic is therefore favored more often than not over a full decade, but it is not a safe certainty for this specific rate-and-tax combination

Confidence: MEDIUM — GT-7?, GT-8?, and GT-10? are each reported-by-delegate/unverified; GT-7?'s and GT-10?'s verification paths are named in chain C2, and GT-8?'s verification path is opening a primary decade-by-decade total-return table (e.g., a published annual-returns spreadsheet) and reading the 2000-2009 annualized figure directly rather than via a secondary synthesis.

Conclusion C5: The pure dollar comparison in C1-C4 understates Option A's value and overstates Option B's liquidity advantage specifically in adverse macro states

C1 (illiquid, guaranteed $111,900) + C2 (liquid, historical-average $133,600) + C3 (liquid, conservative bracket $68,900-$96,200) → the $60,000 committed to Option A is recoverable only via sale, refinance, or a HELOC, each carrying real friction: typical refinance closing costs of roughly 2%-5% of the loan amount, and HELOC rates often priced at or above the original 6.25% note rate [2nd-order, actor lens] → that friction is not evenly distributed over time: lenders tighten HELOC and cash-out-refinance underwriting during recessions and job-loss cycles [2nd-order, time lens; [Assumes: A-5]] → those same cycles are when a household is most likely to want the cash, so Option A's illiquidity cost concentrates in exactly the states where liquidity is most needed [3rd-order, actor lens] → Option B's position is sellable at any point within the 10 years subject only

to the 18% exit tax already priced into chains C2 and C3, so it carries no equivalent state-contingent illiquidity cost [2nd-order, time lens] → the same extra principal payment also lowers loan-to-value and total remaining debt service, reducing exposure to a home-price decline, a protective effect the pure dollar comparison does not capture [2nd-order, actor lens] → on net, the dollar comparison in C1-C4 understates Option A's value and overstates Option B's liquidity advantage in adverse macro states, while in calm states Option B's liquidity advantage is real but less needed

Confidence: MEDIUM — this chain cites C2 and C3, both MEDIUM (their own confidence lines carry the explanation); its own added downgrade cause is the [Assumes: A-5] premise on the second hop, whose verification path is historical HELOC/cash-out-refinance origination-volume data by recession period (e.g., HMDA data); if that premise fails, the illiquidity cost would be less state-correlated but would not disappear, so the qualitative direction of this chain's conclusion is not reversed by it.

Conclusion C6: The weighted trade-off recommends allocating the $60,000 predominantly to mortgage paydown, with an 80/20 paydown/invest split as a defensible liquidity-preserving alternative

C1 (111,900) + C2 (133,600) + C3 (68,900-96,200 bracket) + C5 (liquidity/behavioral second-order effects) → weighting guaranteed-return capture, historical-average expected value, conservative-scenario downside protection, liquidity/flexibility, diversification, and behavioral debt-aversion value — weights locked before scoring, with downside protection weighted highest given the unusually high, untaxed, risk-free mortgage rate — produces locked, pre-scored weighted totals of 70 for 100% paydown, 60 for a 60/40 paydown/invest split, and 50 for 100% invest [Assumes: A-7, A-8, A-10] → that ranking is robust: no single criterion's weight change within the procedure's 1-5 scale reverses either comparison, and the paydown-vs-invest margin stays at least 4 points even at each criterion's floor or ceiling value → an 80/20 paydown/invest split scores 66, within 4 points of pure paydown's 70, while converting roughly $12,000 of the $60,000 into a genuinely liquid, growth-exposed sleeve → the trade-off therefore recommends allocating the $60,000 predominantly to mortgage-principal paydown, with an 80/20 paydown/invest split as a defensible, lower-regret conditional alternative

Confidence: MEDIUM — this chain cites C2, C3, and C5, each MEDIUM (their own confidence lines carry the explanation); its own added downgrade cause is the [Assumes: A-10] premise on the first hop — the weight placed on downside protection is a risk-preference choice, and no empirical observation settles the "correct" weight, so no verification path exists for this cause beyond disclosing it, which is the stated reason in place of a verification path; A-7 and A-8 are lower-weight criteria (diversification, behavioral value) whose failure would shift the totals only slightly and would not change the ranking given the robustness margin in the second hop. This A-10 disclosure is named explicitly as the main counterargument to the headline recommendation in section 6.

5. Abandoned Reasoning

Dead End: Recommend 100% invest in equities (Option B) as the headline answer, on the "equities always beat an un-sheltered mortgage" convention

What was tried: Treated the generic personal-finance heuristic — surplus cash should always be invested rather than used to prepay a mortgage, because equities historically outperform mortgage rates — as sufficient justification on its own, given the household's long 10-year horizon.

Why abandoned: The breakeven analysis (chain C4) shows the required after-tax equity return to merely match the guaranteed paydown outcome is ≈7.8%/yr nominal — close enough to the long-run historical average (GT-7?, ≈9.9%/yr) that the margin of safety is thin, and the conservative-scenario bracket (chain C3) and the realized 2000-2009 "lost decade" (GT-8?, ≈-0.95%/yr) both show Option B losing to Option A by $30,000-$57,000 under plausible 10-year outcomes. The formal weighted trade-off (chain C6) also ranks 100%-invest last, robustly, under locked pre-specified weights. The heuristic is not false in general, but it is not load-bearing enough on its own to be the analysis's answer at this specific rate (6.25%, no tax shield).

What it ruled out: A default "just invest it" recommendation that would have ignored the unusually high, untaxed, risk-free hurdle rate this specific mortgage represents, and would have understated the real probability (not merely the historical average) of a below-breakeven decade.

Dead End: Model a mortgage recast or cash-out refinance as part of Option A

What was tried: Considered incorporating a recast (lowering the monthly payment after the extra principal payment) or treating the paid-down equity as recoverable via a future cash-out refinance within the 10-year window, to give Option A some of Option B's liquidity.

Why abandoned: The problem statement explicitly stipulates "no recast requested/assumed" and that the $60,000 becomes illiquid "unless they sell/refinance/recast later" — modeling a recast or an assumed future refinance would answer a different question than the one posed, and would require additional unstipulated assumptions (recast fees, future rate environment, future qualification) with no anchoring ground truth.

What it ruled out: A rosier Option A variant that assumes away its central liquidity cost, which would have made the comparison in chain C5 meaningless.

Dead End: Build exact lot-level cost-basis tracking for Option B's reinvested dividends

What was tried: Attempted to track each year's after-tax dividend reinvestment as its own tax lot, to compute the exact capital-gains tax at sale rather than approximating the entire gain above the original $60,000 as taxable.

Why abandoned: This requires reinvestment dates and amounts that do not exist as stipulated facts in this problem, and the exact tracking would only ever raise Option B's after-tax outcome slightly relative to the chains C2/C3 approximation (since some of the ending value is already-taxed reinvested-dividend principal, not fresh gain). The approximation's bias direction is known and conservative against Option B, which is disclosed explicitly as [Assumes: A-6] rather than silently assumed, so the simplification does not change the qualitative conclusion.

What it ruled out: A more precise but data-unavailable tax model that would not have changed which option wins under any of the three return scenarios.

6. Conclusion

Recommended approach: Apply the full $60,000 to extra mortgage principal (Option A) as the headline recommendation, because it is the weighted trade-off's robust winner (chain C6); a household that places outsized subjective value on liquidity or optionality beyond what the trade-off's weights capture may instead choose an 80/20 paydown/invest split ($48,000 principal / $12,000 invested), which gives up only about 4 of 70 weighted-score points while converting roughly $12,000 into a genuinely liquid, growth-exposed sleeve (chain C6).

Key insight: The conventional "always invest, equities beat the mortgage" heuristic is not obviously wrong here, but it is thinner than it looks: the required after-tax equity return just to match the guaranteed payoff is about 7.8%/yr nominal, which sits below the long-run historical average of about 9.9%/yr but above a conservative forward-return bracket (about 2%-6%/yr) and above at least one real realized 10-year window, the 2000-2009 "lost decade" at about −0.95%/yr (chain C4). Because this specific mortgage carries an unusually high, fully untaxed, guaranteed 6.25% hurdle rate — higher than a typical post-2008 mortgage and with no deduction to soften it — the margin by which equities need to win is narrower than the generic heuristic assumes (chain C1, chain C4).

Trade-offs acknowledged: This recommendation gives up full liquidity on the committed funds — recoverable only by sale, refinance, or a HELOC, each carrying real cost and each tending to tighten in availability exactly when a household most needs the cash (chain C5) — and it gives up the roughly $21,700 of additional expected value Option B would have produced if the historical average return actually materializes over this specific decade (chain C2, chain C6). It also rests on a stated risk-capital preference — the weight placed on downside protection in the trade-off — that is a defensible judgment call, not a

measured fact; a household that weighs liquidity and upside participation more heavily than downside protection would reasonably land closer to a 50/50 split or even favor Option B outright (chain C6).

Confidence: MEDIUM — the headline recommendation rests on chain C6, which is itself MEDIUM because it cites C2, C3, and C5 (each MEDIUM on unverified return/liquidity inputs, GT-7?, GT-8?, GT-9?, GT-10?, and the A-5/A-9 liquidity assumptions) and carries its own disclosed preference-sensitivity cause (A-10, the trade-off weighting). The verification that would most directly raise this to HIGH is opening primary sources for GT-7? and GT-8? (a published historical-returns dataset) rather than relying on a secondary web-search synthesis, and substituting a named, cited forward-return forecast for the illustrative GT-9? bracket; even without that verification, the recommendation is robust to single-criterion weight changes within the trade-off's 1-5 scale (chain C6), which is why it is presented as MEDIUM rather than LOW.